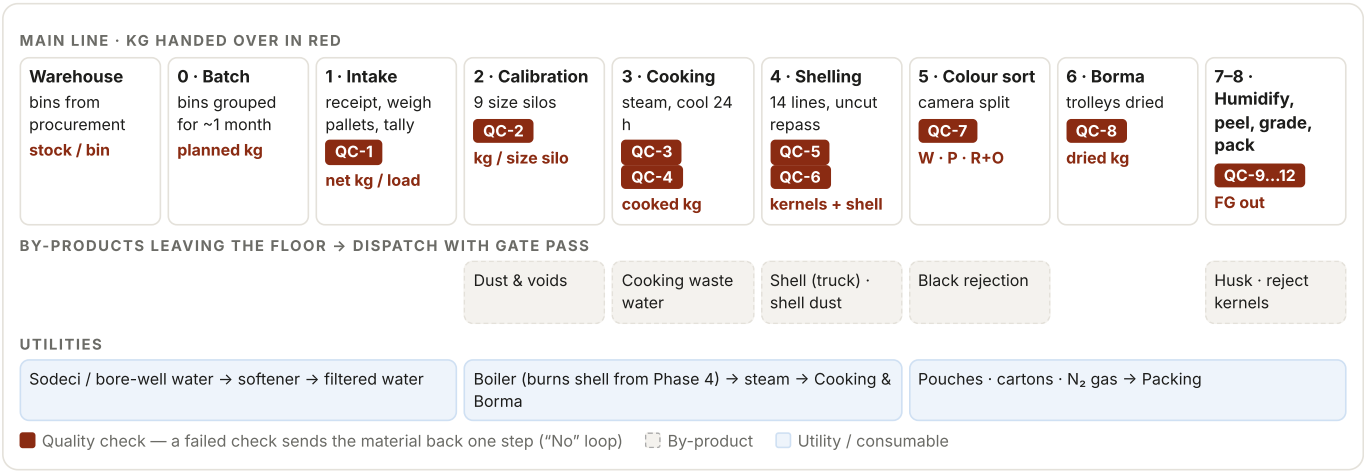
ANAGROCI · RCN TRACKING

Production module

How raw cashew nuts move from the factory warehouse to packed kernels: every process step, every quality check (QC-1 to QC-12), every by-product that leaves the floor, what is recorded at each point, by whom, and what the system calculates.

The flow



Each phase uses what the previous phase produced. The app carries the kg forward, so the next phase starts with the right input and every loss — and every by-product sent out — is visible.

Three levels of tracking

LEVEL	EXAMPLE	WHAT IT IS	USED IN
Batch	ANA-03B	Warehouse bins processed together (≈ 4,000 t, 1 month). Every production record belongs to a batch.	All phases
Load (BTM)	BTM 109	One transfer from one bin to the production floor, with its slip number.	Intake, Calibration
Production Lot	ANG26G16	One cooking batch, created at cooking (ANG · 26 year · G = July · 16th lot of the month). Everything after cooking is tracked by Lot.	Cooking → Dispatch

Who does what

ROLE	RECORDS	APPROVES
Data Entry NEW ROLE	Weights, quantities, trolleys, trucks and gate documents, Excel uploads	—
QFS Officer	All 12 quality checks and the uncut quantity check (moisture, cut tests, machine and shell samples, in-process and final analysis)	—
Warehouse Incharge (Yakro)	Creates batches; can record anything directly	Everything submitted by Data Entry and QFS Officer; closes the batch
Admin	Everything	Unlocks locked records

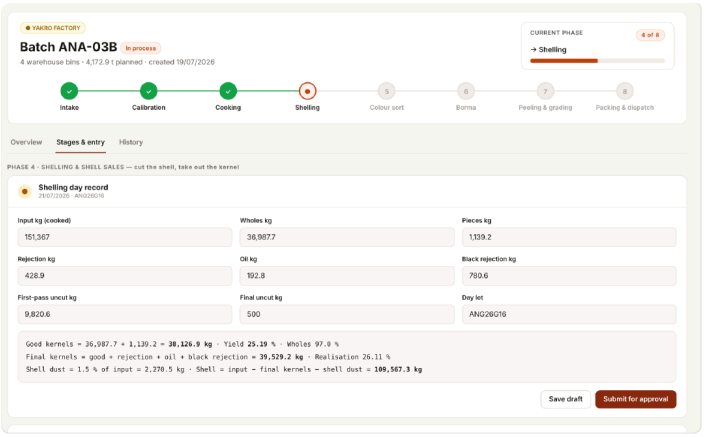
How an entry gets saved

1. **Data Entry / Warehouse Incharge / QFS** — **Fill the form**: Raw values only — no totals or %
2. **App** — **Calculates**: Net weights, yields, losses, % shown live
3. **Data Entry / Warehouse Incharge / QFS** — **Submit**: Saved as pending; nothing changes yet
4. **Warehouse Incharge** — **Approve / reject**: Approve → applied, stock updated. Reject → reason required
5. **App** — **Locked**: Final records reopened only by Admin

OVERVIEW

Where it lives in the app

Sidebar: **Production → Batches**. Everything else happens on the batch page, the same way the procurement Lot page works.



Batch page: header with the 8-phase stepper (done ✓ · current ● · next), tabs Overview · Stages & entry · History, and the entry cards of the current phase.

One batch through the factory — real 2026 figures

PHASE	RECORD	IN	OUT	REMOVED / SENT OUT
0 Batch	ANA-03B	4 bins	4,172.9 t planned	—
1 Intake	BTM 109 · 20/07	174,333.3 kg gross	**167,585.35 kg** net	bags 1,442.35 · pallets 5,305.6
2 Calibration	BTM 113 · 25/07	137,831.9 kg	**136,573.9 kg** into silos	dust 1,250 · voids 8 → waste dispatch
3 Cooking	ANG26G16 · 20/07	from 9 silos	**151,367 kg** cooked	cooking waste water → waste dispatch
4 Shelling	ANG26G16 · 21/07	151,367 kg	**39,529.2 kg** kernels (26.1 %)	shell 109,567.3 (boiler / truck) · shell dust 2,270.5 · uncut 500 repassed
5 Colour sort	ANG26G16 · 21/07	39,529.2 kg	wholes 36,987.7 · pieces 1,139.2 · rej + oil 621.7	black rejection 780.6 → storage → sale
6 Borma	ANG26G16 · 21/07	38,748.6 kg	**36,281.7 kg** dried	moisture 2,466.9 (6.4 %)
7 Humidify · peel · grade	Manual peeling · 17/08	input TW	**74.8 kg** graded (WW 44 · OW 7.5 · SB 5.9 …)	husk → winnowing · rejection 5.6 → sale
8 Packing	Per grade and SKU: packed, QC-9 and QC-11 passed, dispatched, closing stock and aging (Packing Traceability FG Aging Report)			

Example values are from the 2026 production reports for batch ANA-03B; user names on screens are placeholders. Over the whole batch: 38 loads, ≈ 4,173 t received from BIN-03B (24 loads), BIN-10 (9), BIN-16 (4), BIN-15 (1).

OVERVIEW

Quality checks along the line

The flow chart has twelve quality checks and one quantity check. Each one sits between two process steps. When the answer is **No**, the material does not move forward: it goes back to the step shown and the check is repeated. Every check is recorded by the QFS Officer and approved by the Warehouse Incharge.

CHECK	PHASE · POSITION	WHAT IS CHECKED	IF "NO"	REPORT / SHEET
QC-1	1 · after RCN receipt, before weighing	Moisture %, nut count, cut test (KOR)	Quality hold → load returned to **Warehouse**	Input RCN Quality Report · Moisture check
QC-2	2 · on calibrated silos	Correct-size % per silo (≥ 85 %)	Re-sized by hand (wholes manual calibration)	Wholes Manual Calibration
QC-3	3 · right after cooking	Cooker parameters met per size (temp, pressure, time)	Re-cook / hold	Cooking Process Tally Sheet · Cooking Process Report
QC-4	3 · after 24 h in cooling silos	Moisture & temperature	Wait in cooling silo → re-check	Moisture & Temperature Check
QC-5	4 · after shelling machines	Machine frequency, capacity, wholes / uncut / unscoop %	Machine adjusted	Shelling Machine Frequency Report · Machine Capacity Check Sample
QTY-1	4 · on uncut nuts	Uncut quantity (kg)	**Repass** through shelling	Shelling day record (first-pass / final uncut)
QC-6	4 · on shell, before shell silo	Kernel left in shell (≤ 0.5 %)	Alert, line checked	Shell Waste Sample Report
QC-7	5 · after colour sorter	Wholes / brokens / rejection % on wholes, pieces, rejection + oil	Re-sort	Shelled Kernels Quality Report
QC-8	6 · after Borma run	Moisture, scorched, spotted, shrivelled …	Trolley **back into Borma**	Final Output Analysis · Borma reports 1–6
QC-10	7 · after humidification	Moisture before peeling	Back to **humidification**	Moisture · Borma Trolley Tally Sheet
QC-12	7 · after colour sorter / dicing / manual peeling / Taiho	Graded and reject kernels	Back to **grading**	In-process Quality Analysis Report
QC-9	8 · at packing, before filling	Peeled / Borma wholes ready to pack	Back to **packing process**	Peeled QA · Borma Wholes QA · In-process QA
QC-11	8 · on finished goods	Final output analysis	Back to **packing process**	Final Output Quality Analysis Report

QC-9 and QC-11 appear twice in the flow chart — once for Borma wholes packed directly, once for graded kernels from Phase 7. The same forms are used in both lanes.

Utilities and consumables

Not phases of their own, but connected to the line. Each has a consumption record so cost per batch can be read.

STREAM	PATH	FEEDS	RECORDED BY	REPORT
Water	Sodeci / bore-well water → softener → filtered water	Boiler	Data Entry / Warehouse Incharge	Water Consumption
Steam	Boiler (fuel: shell from Phase 4) → steam	Cooking (3) · Borma (6)	Data Entry / Warehouse Incharge	21 Boiler Consumption
Packing material	Pouches · cartons · N_2 gas	Filling into pouch (8)	Data Entry / Warehouse Incharge	N_2 Gas, Pouches & Cartons Consumption

FIELD	ENTERED BY	RULE
Water: date · source (Sodeci / bore well) · meter opening · closing	Data Entry / Warehouse Incharge	m^3 used = closing − opening, per source per day
Boiler: date · shell kg burned · running hours	Data Entry / Warehouse Incharge	Shell kg taken from the shell balance of Phase 4
Packing material: date · item · opening · received · used	Data Entry / Warehouse Incharge	Closing = opening + received − used; used per kg packed shown

PHASE 0

Batch creation

Choose which warehouse bins will be processed next and give the group a name. It is a plan: no stock moves yet.

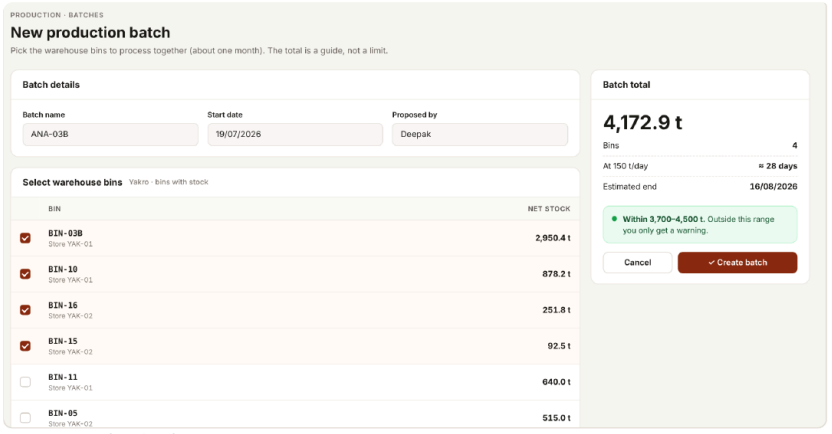
1. Warehouse team — **Propose bins** — e.g. BIN-03B, BIN-10, BIN-16, BIN-15
2. Warehouse Incharge — **Open New batch** — Production → Batches → New batch
3. Warehouse Incharge — **Tick bins** — Only Yakro bins with stock; bins in a running batch greyed out
4. App — **Check total** — Tonnes, days, end date, monthly guide
5. Warehouse Incharge — **Create** — Batch saved as Open

RECEIVES
Warehouse bins and their stock (procurement)

HANDS OVER
The bin list that Intake can draw loads from

FIELD	ENTERED BY	RULE
Batch name	Warehouse Incharge	Required, unique. Suggested: ANA- + largest bin (BIN-03B → ANA-03B)
Start date · Proposed by	Warehouse Incharge	Date defaults to today
Bins	Warehouse Incharge	One or more; a bin can be in only one open batch
Planned kg · days · end date	App	See formulas
Status	App	Open → In process (first load received) → Closed (Phase 8)

```
Planned = Σ net stock of selected bins = 2,950.4 + 878.2 + 251.8 + 92.5 = 4,172.9 t
Days = planned ÷ 150 t/day ≈ 28 · End = start date + days
Guide: 3,700–4,500 t (150 t × 26 days). Outside the range → warning only, never blocked.
```



New batch — the four bins of ANA-03B.

PHASE 1

Intake

Every load taken from a batch bin to the production floor is one RCN Receipt (ANA-03B: 38). The load is quality-checked first, weighed pallet by pallet, tipped into the calibration line, and then the empties, jute bags, dust and voids are weighed so the tally sheet can be reconciled.

1 Data Entry / Warehouse Incharge
RCN receipt
BTM no, date, which bin

→

2 QFS Officer
QC-1 **Moisture check**
Moisture %, nut count, cut test

→

3 Data Entry / Warehouse Incharge
Weigh all pallets
Each pallet: gross kg, bags

→

4 Data Entry / Warehouse Incharge
Process into calibration
Nuts tipped into the line

→

5 Data Entry / Warehouse Incharge
Empties, dust & voids
Pallets, jute bags, dust, voids weighed

→

6 App WI
Reconcile tally
Net and into-silo kg; approved, locked

QC-1 = NO
Load goes on **Quality hold** and is returned to the **Warehouse** bin it came from. Weighing stays closed. The receipt and the failed sample are kept; the bin stock is not reduced.

RECEIVES
A bin of the batch (Phase 0)

HANDS OVER
Net kg of the load and dust/voids → Calibration (Phase 2) · dust & voids → waste dispatch

Step 1 · RCN receipt

FIELD	ENTERED BY	RULE
Batch	Data Entry / Warehouse Incharge	Only Open or In-process batches
Date · BTM no	Data Entry / Warehouse Incharge	BTM from the transfer slip; same BTM + same bin twice → warning
From bin	Data Entry / Warehouse Incharge	Only the batch's bins, with kg left in each
Receipt ID · status	App	RR-YAK-2026-0001 … · status Received. First receipt sets the batch to In process

Step 2 · QC-1 Moisture check

FIELD	ENTERED BY	RULE
Moisture % · nut count per kg	QFS Officer	Moisture 0–20 %
Sample g and grams of: good kernel, spotted, immature, oil, rejection, voids, shell, fruit waste	QFS Officer	Parts must add up to the sample weight (normally 1,000 g)
Decision	QFS Officer	OK → weighing opens. No → Quality hold, returned to warehouse
KOR · SHOT · FOT · yields	App	See formulas

KOR = (good kernel + ½ × (spotted + immature)) × 0.17637 → BTM 109: (243 + ½ × 21.5) × 0.17637 = **44.75**
SHOT = (good + spotted + immature) × 0.17637 · FOT = SHOT parts + oil + rejection, × 0.17637
0.17637 converts grams per 1 kg sample into lbs per 80 kg bag.

PHASE 1 · INTAKE (CONTINUED)

Step 3 · Weigh all pallets (tally sheet)

FIELD	ENTERED BY	RULE
Line no	App	Numbered automatically, one line per pallet
Gross kg	Data Entry / Warehouse Incharge	Pallet + bags + nuts, as read on the scale
Bags	Data Entry / Warehouse Incharge	Usually 30 per pallet
Bags kg	App	bags × 0.65

Steps 4–5 · Process into calibration, then weigh empties, dust & voids

FIELD	ENTERED BY	RULE
Date / time tipped into calibration	Data Entry / Warehouse Incharge	Receipt status → In calibration
Empty pallets kg	Data Entry / Warehouse Incharge	All pallets of the load weighed together after the nuts are tipped
Jute bags kg	Data Entry / Warehouse Incharge	Optional actual weight; if blank the app uses bags × 0.65
Dust kg · voids kg	Data Entry / Warehouse Incharge	Removed by the calibration machine; cannot exceed net. Go to the dust & voids store

Step 6 · Reconciliation of the tally sheet

FIELD	ENTERED BY	RULE
Total gross · total bags · bags kg · net · into silos · dust % · voids %	App	See formula
Approval	Warehouse Incharge	On approval the receipt is locked and appears in Calibration

```
Net = Σ gross − bags kg − empty pallets
BTM 109: 174,333.30 − 1,442.35 − 5,305.60 = 167,585.35 kg (75 pallets, 2,219 bags)
Into silos = net − dust − voids → BTM 113: 137,831.90 − 1,250 − 8 = 136,573.90 kg (dust 0.91 %, voids 0.006 %)
```

> **BY-PRODUCT · DUST & VOIDS**
> Stored, then sent out by truck through the common gate procedure (see *By-products & dispatch*). Recorded in the **Waste Management Dispatch Tracker**.

What updates on approval

WHERE	CHANGE
Batch	Received +167,585.35 kg; bin progress bar moves
Warehouse bin BIN-03B (procurement)	Stock −167,585.35 kg
Procurement Lot page (Lots in BIN-03B)	Shows “Sent to production · ANA-03B”
Production RCN stock	Opening + received − sent to calibration = closing
Dust & voids store	+ dust kg + voids kg (waiting for dispatch)

Reports produced

RCN Receipts Tally Sheet · Input RCN Quality Report · RCN Stock Report (calculated).

Receipt status

Received → Quality checked (or **Quality hold → returned to warehouse**) → Weighed → In calibration → Reconciled (locked) → Calibrated (Phase 2). Statuses are worked out from the records, never typed.

PHASE 2

Calibration

The reconciled load runs through the sizing machine into 9 storage silos (A++ largest … E smallest). The QFS Officer checks that each silo holds the right size.

1 Data Entry / Warehouse Incharge
Pick reconciled load
Only locked receipts not yet calibrated

→

2 Data Entry / Warehouse Incharge
Kg into each silo
Daily, 9 sizes

→

3 QFS Officer
QC-2 Sizing check
Sample per silo

→

4 Data Entry / Warehouse Incharge
Wholes manual calibration
Only for silos that fail

→

5 App
Silo stock
Per size, always current

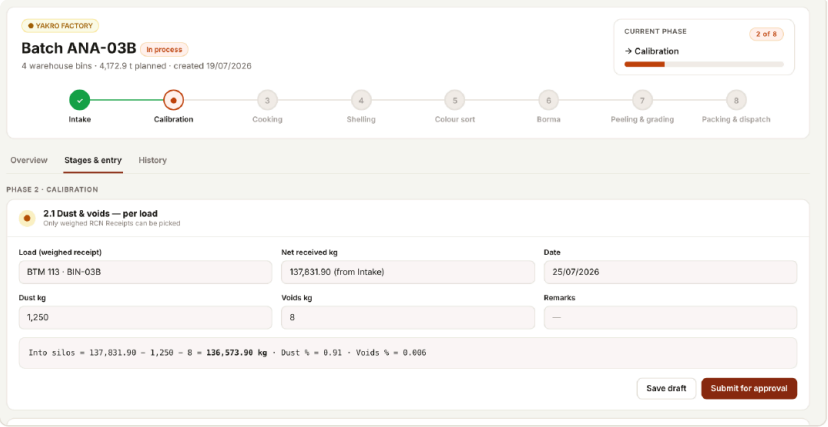
QC-2 = NO (CORRECT SIZE BELOW 85 %)
Silo is flagged; nuts are re-sized by hand. The kg moved between sizes is entered as a **Wholes Manual Calibration** line and adjusts both silos.

RECEIVES
Into-silo kg of a reconciled RCN Receipt (Phase 1)

HANDS OVER
Kg in each size silo → Cooking (Phase 3)

FIELD	ENTERED BY	RULE
RCN Receipt · into-silo kg	Data Entry / Warehouse Incharge App	Copied from the reconciled receipt
Kg per size: A++ · A+ · A · AB · B · BC · C · D · E	Data Entry / Warehouse Incharge	Daily; the 9 sizes should add up to the kg put into silos that day
Sizing check: silo, nut count, grams found in each size	QFS Officer	1,000 g sample; alert if less than 85 % is the correct size
Manual calibration: from size, to size, kg	Data Entry / Warehouse Incharge	Only after a failed QC-2; kg moves between silos
Silo stock per size	App	See formulas

```
Silo stock (each size) = opening + sized in − cooked ± manual calibration
Correct size % = grams of the silo's own size ÷ sample
```



Calibration on the batch page — BTM 113. (Dust & voids are now captured in Intake step 5; the card shows them read-only.)

Reports produced

4 Calibration Process Report · 5 Calibrated RCN Silos Stock Report · 24 Wholes Manual Calibration.

PHASE 3

Cooking & cooling

Each size is steam-cooked separately (1-tonne cookers) so the shell breaks cleanly, checked, then moved to cooling silos for 24 hours and checked again. Steam comes from the boiler, which burns shell from Phase 4.

1 Data Entry / Warehouse Incharge
Cook by size
Kg taken from each silo; cooker log

→

2 QFS Officer
QC-3 **Cooking check**
Parameters met per size

→

3 Data Entry / Warehouse Incharge
To cooling silos
Start of 24 h rest

→

4 QFS Officer
QC-4 **Moisture & temp.**
After 24 h: OK / wait

→

5 App
Ready for shelling
Cooked kg = next day's input

QC-3 = NO
Size is held and re-cooked; the cooker log keeps both runs.

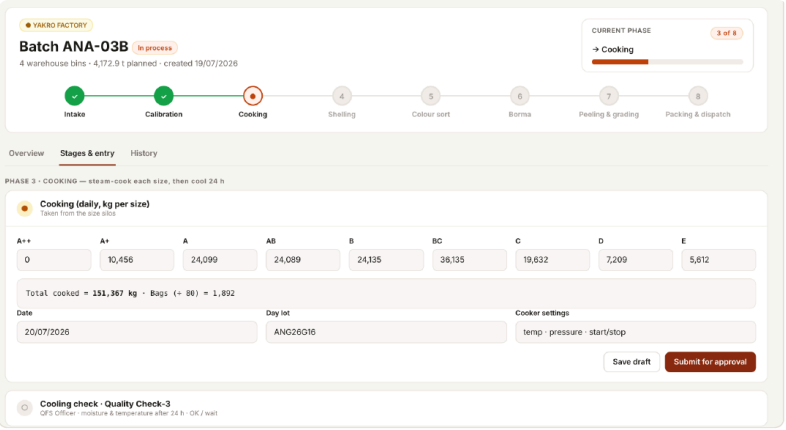
QC-4 = NO
Lot stays in the cooling silo (**Wait**) and is re-checked; shelling cannot start on it.

RECEIVES
Kg in each size silo (Phase 2) · steam from boiler

HANDS OVER
Cooled kg of the Lot → Shelling (4) · cooking waste water → waste dispatch

FIELD	ENTERED BY	RULE
Date · Lot	Data Entry / Warehouse Incharge	Lot e.g. ANG26G16 — created here
Kg cooked per size (9)	Data Entry / Warehouse Incharge	Cannot exceed that silo's stock
Total cooked · bags	App	See formula
Cooker log: size, start, stop, temperature °C, pressure bar	Data Entry / Warehouse Incharge	Limits per size to be agreed
QC-3 cooking check: size, parameters OK, remarks	QFS Officer	No → re-cook
Cooling silo · time in	Data Entry / Warehouse Incharge	24 h clock starts
QC-4 cooling check: silo, moisture %, temperature °C	QFS Officer	Warning if less than 24 h since cooking; Wait → re-check
Cooking waste water: date, kg / litres, to tank	Data Entry / Warehouse Incharge	Adds to waste store; leaves by truck (Waste Management Dispatch Tracker)

```
Total cooked = Σ kg of the 9 sizes → 20/07, ANG26G16: 151,367 kg · Bags = total ÷ 80
Silo stock of each size −= kg cooked of that size
```



Cooking on the batch page — sizes cooked on 20/07/2026; QC-3 and QC-4 cards follow.

Reports: 6 Cooking Process Tally Sheet / Report · Moisture & Temperature Check · 21 Boiler Consumption.

PHASE 4

Shelling, shell & shell sales

14 cutting lines open the shells. Shelling gives four streams: shelled kernels go to the colour sorter; uncut nuts are counted and repassed; shell is checked and stored in the shell silo for the boiler or truck sale; shell dust is sent out as waste.

1 Data Entry / Warehouse Incharge
Day record
Input and every output kg

→

2 Data Entry / Warehouse Incharge
Line runs
Per line: input, times, uncut

→

3 QFS Officer
QC-5 Machine checks
Frequency, capacity sample

→

4 QFS Officer
QTY-1 Uncut
Uncut kg → repass

→

5 QFS Officer
QC-6 Shell sample
Kernel left in shell

→

6 Data Entry / Warehouse Incharge
Shell to silo · trucks
Stock, boiler, sale

UNCUT → REPASS
Uncut nuts go back into shelling. Both first-pass and final uncut are recorded; only *final* uncut leaves the day's balance.

QC-6 ABOVE 0.5 %
Alert on the line; shell still moves to the shell silo, the line is checked (21/07 line 2: 0.78 %).

RECEIVES
Cooled kg of the Lot after QC-4 (Phase 3)

HANDS OVER
Kernels → Colour sort (5) · shell → shell silo → boiler / sale · shell dust → waste dispatch

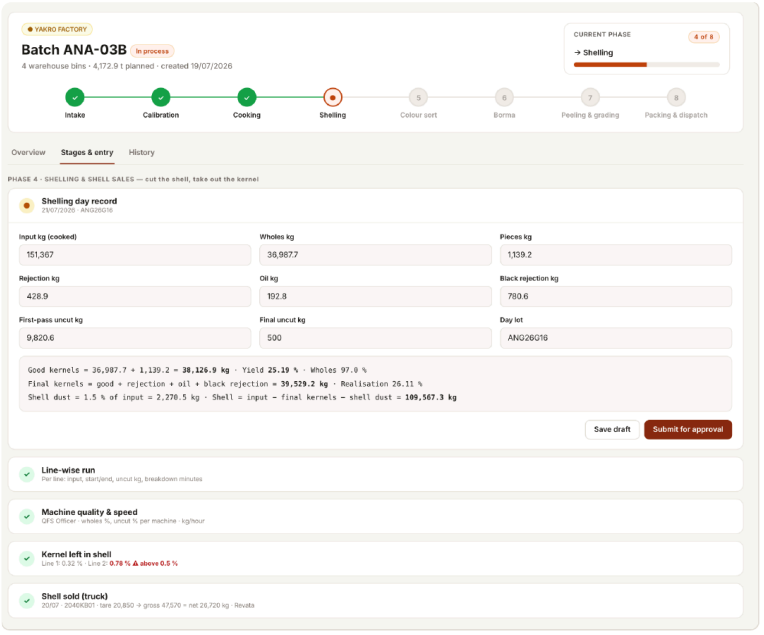
Day record

FIELD	ENTERED BY	RULE
Date · Lot · input kg	Data Entry / Warehouse Incharge	Input pre-filled with the cooled kg that passed QC-4
Wholes · pieces · rejection · oil · black rejection (kg)	Data Entry / Warehouse Incharge	Outputs of the colour sorter after shelling
First-pass uncut · final uncut (kg)	Data Entry / Warehouse Incharge	First-pass uncut is repassed
Good kernels · final kernels · shell dust · shell · all %	App	See formulas — shell is calculated, not weighed

```
Good kernels = wholes + pieces · Final kernels = good + rejection + oil + black rejection
Shell dust = 1.5 % × input · Shell = input − final kernels − shell dust
Yield % = good ÷ input · Realisation % = final ÷ input · Wholes % = wholes ÷ good · Uncut % = first-pass uncut ÷ input
21/07: 151,367 in → good 38,126.9 (25.19 %) · final 39,529.2 · shell 109,567.3 · wholes 97.0 %
```

Checks on the line

FIELD	ENTERED BY	RULE
Line run: line, size, input kg, start, end, uncut kg	Data Entry / Warehouse Incharge	Duration = end − start · uncut % = uncut ÷ input
QC-5 machine frequency: line, machine, grams per minute	QFS Officer	kg per hour = g/min × 0.06
QC-5 capacity sample: shift, line, machine, sample g, wholes, pieces, butts, knife mark, uncut, unscoop, shell (g)	QFS Officer	Wholes % (target 95 %) · uncut % · unscoop %
QTY-1 uncut quantity check: line, kg, repassed Y/N	QFS Officer	Uncut is repassed until final uncut is recorded
QC-6 shell sample: point, sample g, whole, brokens, unscoop, uncut, shell (g)	QFS Officer	Alert if kernel left in shell > 0.5 %



Shelling on the batch page — day record for 21/07/2026 with calculated results, line checks and shell truck.

Shell silo and shell truck

After QC-6 the shell is transferred to the shell silo. From there it is either burned in the boiler (Utilities) or sold. A sale follows the common gate procedure:

Truck arrives at gate → Check documents → Weigh truck (tare) → Position under silo → Load shell → Weigh truck (gross) → Documentation → Sales / dispatch · Non-Returnable Gate Pass / Sortie

FIELD	ENTERED BY	RULE
Shell to silo: date, kg	App	= calculated shell of the day record
Truck: date, truck no, driver, documents checked ✓	Data Entry / Warehouse Incharge	Weighing opens only after documents are checked
Tare kg · gross kg · weighbridge slip	Data Entry / Warehouse Incharge	Net = gross − tare
Buyer · price · Gate Pass / Sortie no	Data Entry / Warehouse Incharge	Gate pass no required to close the truck

Shell silo balance = shell produced − burned in boiler − sold
20/07 · truck 2040KB01 · tare 20,850 → gross 47,570 = **net 26,720 kg** · buyer Revata

> **BY-PRODUCT · SHELL DUST**
> 1.5 % of input, stored and sent out by truck. Waste Management Dispatch Tracker + Non-Returnable Gate Pass.

Reports: 7 Shelling Process · 23 Line-wise · Shelling Machine Frequency · Machine Capacity Check Sample (13) · 11 Shell Waste Sample · 16 Shell Dispatch.

PHASE 5

Colour sorting

A camera machine splits the shelled kernels into wholes, pieces, rejection + oil and black rejection. The kg are already in the shelling day record; this phase records how well the sorter worked (QC-7) and sends black rejection to storage for sale.

1 QFS Officer
Sample before sorter
1,000 g

→

2 QFS Officer
QC-7 **After sorter**
Wholes, pieces, rejection + oil

→

3 App
Compare
% per defect, before vs after

→

4 Data Entry / Warehouse Incharge
Black rejection
To storage

→

5 Warehouse Incharge
Approve

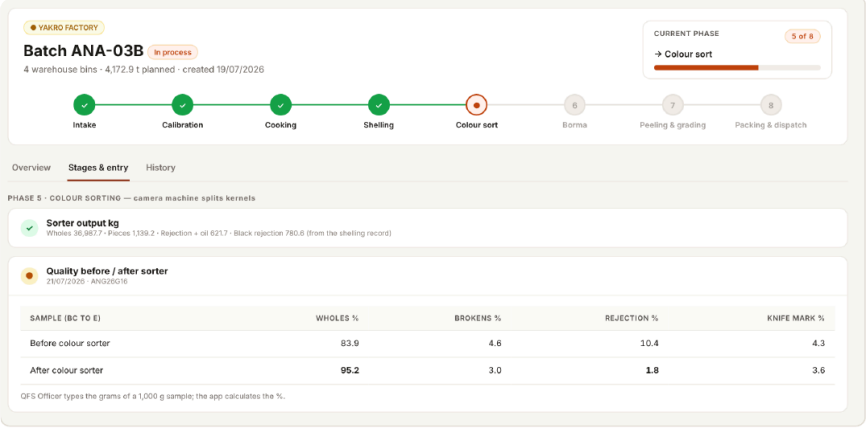
QC-7 = NO
Stream is passed through the sorter again before it goes to the Borma hopper.

RECEIVES
Shelled kernels (Phase 4)

HANDS OVER
Wholes, pieces, rejection + oil → Borma hopper (6) · black rejection → storage → sale

FIELD	ENTERED BY	RULE
Date · Lot · shift	QFS Officer	
Sampling point · stream · size group	QFS Officer	Before sorter / after sorter · wholes / pieces / rejection + oil · A++ to B / BC to E
Sample g · nut count · moisture %	QFS Officer	
Wholes · brokens · butts · knife mark · rejection · oil/dirt · edge cut · shell (g)	QFS Officer	Parts must add up to the sample
All %	App	See formulas
Black rejection to storage (kg) · storage stock	App	From the shelling day record; reduced by sales trucks

```
Rejection % = (rejection + oil/dirt) ÷ sample · Brokens % = (brokens + butts) ÷ sample · Shell % = shell ÷ sample
Wholes % = 100 − rejection % − brokens % − shell %
21/07 (BC to E): before sorter wholes 83.9 %, rejection 10.4 % → after 95.2 %, 1.8 %
```



Colour sort on the batch page — before / after samples for 21/07/2026.

Reports: 12 Shelled Kernels Quality Report · Black rejection sales: Non-Returnable Gate Pass / Sortie.

PHASE 6

Borma (drying)

Kernels are loaded through a hopper into trays on numbered trolleys, weighed, and dried in Borma ovens heated by steam. After QC-8 each trolley either goes back for another run, or moves on — wholes to packing or humidification; pieces and rejection + oil to humidification.

1. Data Entry / Warehouse Incharge — **Hopper · empty trolley** — Trolley no, empty kg
2. Data Entry / Warehouse Incharge — **Trays · gross trolley** — Kernels + trolley kg; tally sheet
3. Data Entry / Warehouse Incharge — **Load · set parameters** — Oven no, steam, 4 cycles
4. Data Entry / Warehouse Incharge — **Record · weigh after** — Run log, gross after
5. QFS Officer — QC-8 — **Quality after Borma** — OK / back to Borma
6. Data Entry / Warehouse Incharge — **Send on** — Packing or humidification

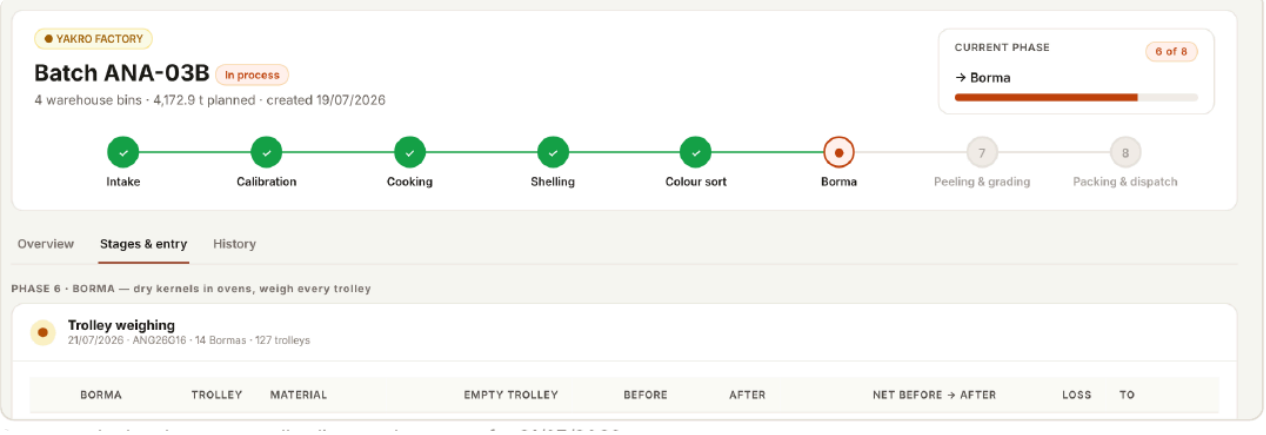
QC-8 = NO
Trolley goes **back into the Borma** (Loading into Borma). A second oven run is logged on the same trolley line; loss % is the total over both runs.

RECEIVES
Wholes, pieces, rejection + oil after QC-7 (Phase 5) · steam

HANDS OVER
Wholes → Packing (8a) or Humidification (7) · pieces, rejection + oil → Humidification (7). In 2026: 6,242 trolley rows to packing, 698 onward

FIELD	ENTERED BY	RULE
Date · Lot · Borma no · trolley no · material	Data Entry / Warehouse Incharge	Material: wholes / pieces / rejection + oil. Empty weight pre-filled from the trolley list
Gross before · gross after (kg)	Data Entry / Warehouse Incharge	After cannot be more than before
Oven run: start, stop, steam pressure bar, temperature and hours for cycles 1–4	Data Entry / Warehouse Incharge	Running hours calculated; a re-run adds a second run
QC-8 quality after Borma: moisture %, good kernels, scorched, spotted, shrivelled, blemish, rejection, husk … (g) · decision	QFS Officer	Parts add up to the sample. No → back to Borma
Goes to	Data Entry / Warehouse Incharge	Wholes: Packing or Humidification · pieces / rejection + oil: Humidification only
Net before · net after · loss · loss %	App	See formulas

```
Net = gross − empty trolley · Loss = net before − net after · Loss % = loss ÷ net before
Trolley 200: 221.4 → 205.2 kg, loss 16.2 kg (7.3 %) · Lot 21/07: 38,748.6 → 36,281.7 kg, loss 6.4 %
The daily Borma summary is built from the trolley lines — no separate entry.
```



Borma on the batch page — trolley lines and oven run for 21/07/2026.

Reports: Borma temperature, pressure, time · Moisture · 18 Borma Trolley Tally Sheet · 20 Borma Machine Running Hours · 8 Borma Process Report (calculated) · 14 Final Output Analysis.

PHASE 7

Humidification, peeling & grading

Kernels routed from the Borma are humidified so the skin (testa) comes off, peeled, and graded into the grades they are sold under. The husk is winnowed to recover pieces. Four sub-steps, two quality checks.

7a Data Entry / Warehouse Incharge
Humidification
Trolleys in, kg out

→ QFS Officer
QC-10
Moisture
OK / back

→ 7b Data Entry / Warehouse Incharge
Peeling process
Wholes · pieces · husk

→ 7c Data Entry / Warehouse Incharge
Husk winnowing
Recovered pieces · husk

→ 7d Data Entry / Warehouse Incharge
Colour sort · dicing · manual peeling · Taiho
Graded + reject

→ QFS Officer
QC-12 **In-process**
OK → packing

QC-10 = NO · QC-12 = NO
Trolley goes back to **humidification** · kernels go back to **grading** (7d).

RECEIVES
Borma trolleys routed to Humidification (Phase 6)

HANDS OVER
Graded kernels → Packing (8b) · reject kernels → sale · husk → storage → sale

7a · Humidification and QC-10

FIELD	ENTERED BY	RULE
Date · Lot · chamber · trolley nos · kg in	Data Entry / Warehouse Incharge	Only trolleys routed to Humidification; kg in = Borma net after
Start · end · kg out	Data Entry / Warehouse Incharge	Moisture gain = kg out − kg in
QC-10: moisture %, decision	QFS Officer	No → back to humidification. Sheets: Moisture · Borma Trolley Tally Sheet

7b · Peeling process and 7c · Husk winnowing

FIELD	ENTERED BY	RULE
Peeling: date, Lot, machine, kg in, wholes kg, pieces kg, husk kg	Data Entry / Warehouse Incharge	Outputs ≤ kg in; machine quality sample by QFS (Machine Quality Sample Report)
Winnowing: husk kg in, pieces recovered kg, husk to storage kg	Data Entry / Warehouse Incharge	Recovered pieces join 7d; husk to storage → sale

7d · Grading: colour sorter, dicing, manual peeling & Taiho

Today only the **manual peeling** route has a workbook: kg out per grade and the workers used. The other routes use the same grade list with a route field.

FIELD	ENTERED BY	RULE
Date · unit / route · input material	Data Entry / Warehouse Incharge	Route: colour sorter / dicing / manual peeling / Taiho. Input as written in the sheet (e.g. TW)
Kg per grade	Data Entry / Warehouse Incharge	Codes kept exactly as the sheet: WW, CW, SW, SSW, DW, PUK1–3, SS, LP, DP, OW, DW2, FB, SB, FS, LWP, OP, SP, SSP2, KTP, SP2, Powder
Reject kernels kg	Data Entry / Warehouse Incharge	"Rejection" column of the sheet → reject store → sale
No of manpower · worker output (employee no, name, kg per day)	Data Entry / Warehouse Incharge	Manual peeling only; one line per worker present
QC-12 in-process analysis: graded and reject samples	QFS Officer	No → back to grading. In-process Quality Analysis Report
Total · avg per head · grade share · band	App	See formulas
Recoverability %	Data Entry / Warehouse Incharge	Column exists in the sheet; empty in 2026 rows — rule to be agreed

Total kg = Σ grades + reject · Avg per head = total ÷ manpower · Grade share % = grade kg ÷ total
17/08/2026: WW 44 + CW 4 + SW 0.8 + DW 1.6 + PUK2 2.1 + PUK3 0.9 + SS 1 + LP 1.4 + OW 7.5 + SB 5.9 + Rejection 5.6 = **74.8 kg** · 11 workers → **6.8 kg/head**
Worker band (kg/day): GOOD > 11 · AVERAGE 9–11 · NEED IMPROVEMENT 7–9 · POOR 5–7 · V.POOR < 5

PHASE 8

Packing, finished goods & closing the batch

Two lanes reach packing: **8a** Borma wholes sent straight from Phase 6, and **8b** graded kernels from Phase 7. Both follow the same steps and the same two checks.

1. Data Entry / Warehouse Incharge — **Packing process** — Grade, SKU, Lot, bin
2. QFS Officer — **QC-9 Before filling** — OK / back to packing
3. Data Entry / Warehouse Incharge — **Fill into pouch** — Pouches, cartons, N_2 used
4. App — **Finished goods** — Stock and aging
5. QFS Officer — **QC-11 Final output** — OK / back to packing
6. Data Entry / Warehouse Incharge — **Dispatch** — Contract, gate pass

QC-9 = NO
Material goes back to the **packing process** (re-sort / re-blend); nothing is filled.

QC-11 = NO
Packed goods are held and go back to the **packing process**; FG stock is not released for dispatch.

RECEIVES
Borma wholes (6) and graded kernels after QC-12 (7) · pouches, cartons, N_2

HANDS OVER
Finished goods to customers · batch closed

FIELD	ENTERED BY	RULE
Date · lane (8a / 8b) · grade · material code · SKU	Data Entry / Warehouse Incharge	SKU = sellable product (grade + pack)
Lot · bin	Data Entry / Warehouse Incharge	Links the pack back to production
QC-9 pre-fill analysis	QFS Officer	Peeled Quality Analysis (8b) · Borma Wholes Quality Analysis (8a) · In-process QA
Quantity packed (pieces, kg) · pouches · cartons · N_2 used	Data Entry / Warehouse Incharge	Consumables reduce the packing-material stock (Utilities)
QC-11 final output analysis	QFS Officer	Same sample format as QC-8 (Final Output Quality Analysis Report). FG released only when OK
Dispatch: contract no, quantity out (pieces, kg), truck, Gate Pass / Sortie no	Data Entry / Warehouse Incharge	Dispatch against a sales contract; only QC-11-passed stock
Closing quantity · status · aging (days)	App	See formulas

```
Closing quantity = packed − dispatched (pieces and kg) · Aging = today − packing date
Lot status: … → Packed → QC-11 passed → Dispatched once any of its packed quantity has shipped
```

Closing the batch

The Warehouse Incharge closes the batch when all its Lots are packed. The app shows every stream that left the line, so the balance can be checked before locking.

```
Batch balance = RCN received
  − dust − voids − shell − shell dust − final uncut − black rejection
  − Borma moisture loss + humidification moisture gain − husk − reject kernels
  → compared with kg packed; difference shown as unaccounted loss (alert above tolerance, rule to be agreed)
```

Cooking adds water and is not used in the kernel balance; cooking waste water is tracked as a waste stream only.

Reports: Packing Traceability FG Aging Report · Final Output Quality Analysis · N_2 Gas, Pouches & Cartons Consumption.

REFERENCE

By-products & dispatch

Everything that leaves the factory by truck — by-products, waste and finished goods — goes through the same gate procedure and gets a Non-Returnable Gate Pass / Sortie. One form, with the stream picked from a list.

Truck arrives at gate → Check documents → Weigh truck (tare) → Position / load → Weigh truck (gross) → Documentation → Sales / dispatch · Gate Pass / Sortie

STREAM	PRODUCED IN	HELD IN	LEAVES AS	REGISTER
Dust & voids	1 Intake / calibration line	Dust & voids store	Waste dispatch	Waste Management Dispatch Tracker
Cooking waste water	3 Cooking	Waste water tank	Waste dispatch	Waste Management Dispatch Tracker
Shell	4 Shelling (after QC-6)	Shell silo	Boiler fuel or sale	16 Shell Dispatch · Gate Pass / Sortie
Shell dust	4 Shelling	Shell dust store	Waste dispatch	Waste Management Dispatch Tracker · Gate Pass
Black rejection	5 Colour sort	Storage	Sale	Gate Pass / Sortie
Husk	7c Husk winnowing	Storage	Sale	Gate Pass / Sortie
Reject kernels	7d Grading	Reject store	Sale	Gate Pass / Sortie
Finished goods	8 Packing (after QC-11)	FG store	Sale against contract	FG Aging Report · Gate Pass / Sortie

FIELD	ENTERED BY	RULE
Date · stream · truck no · driver · documents checked ✓	Data Entry / Warehouse Incharge	Weighing opens only after documents are checked
Tare kg · gross kg · weighbridge slip	Data Entry / Warehouse Incharge	Net = gross − tare; net cannot exceed the stream's store stock
Buyer / vendor · category · price · contract no (FG)	Data Entry / Warehouse Incharge	Contract required for finished goods
Gate Pass / Sortie no	Data Entry / Warehouse Incharge	Required to close the truck; unique
Approval	Warehouse Incharge	On approval the store stock is reduced

`Store balance (each stream) = produced − dispatched` (shell: − burned in boiler)

REFERENCE

Statuses, data and imports

Statuses

RECORD	STATUSES
Batch	Open → In process (first load) → Closed (by Warehouse Incharge)
BTM (RCN Receipt)	Received → Quality checked / Quality hold (returned) → Weighed → In calibration → Reconciled → Calibrated
Production Lot	Cooking → Cooling → Shelled → Colour sorted → In Borma → Humidified → Peeled → Graded → Packed → Dispatched (wholes packed directly skip Humidified · Peeled · Graded)
Quality check	OK · No (material returned to the step shown in the QC map) · Wait (QC-4)
Daily records, samples, trucks	Draft → Submitted → Approved (locked) / Rejected (reason kept)

Main data records

RECORD	ONE ROW =	LINKED TO
Batch · Batch bin	a batch · a bin in it	procurement warehouse bin
RCN Receipt · Pallet line	a load · a pallet (incl. empties, bags, dust, voids)	batch, bin
Silo movement	kg per size per day (sized in, cooked, manual calibration)	receipt, batch
Lot	one day of processing	batch
Cooking · Shelling · Humidification · Peeling · Grading · Packing day	one Lot / day each	Lot
Line run · Machine check · Trolley line · Oven run	one line / machine / trolley / oven run	Lot, equipment
Quality check	one sample, with check no (QC-1…12, QTY-1) and decision	receipt or Lot
By-product store · Truck movement	one stream balance · one truck out with gate pass	day / batch
Utility reading	water, boiler, packing material per day	day / batch

Masters: sizes (9), lines (14), machines, cookers, cooling silos, Borma ovens, trolleys with empty weight, humidification chambers, grades, by-product streams, vendors / buyers. Settings: bag weight 0.65 kg, daily capacity 150 t, monthly guide 3,700–4,500 t.

Alerts

Batch total outside 3,700–4,500 t · QC-1 moisture fail (quality hold) · sample parts ≠ sample weight · correct-size % below 85 (QC-2) · cooking parameters out of limit (QC-3) · cooling under 24 h (QC-4) · wholes below 95 % (QC-5) · kernel in shell above 0.5 % (QC-6) · Borma after weight higher than before · trolley re-run (QC-8 No) · any QC returned twice · worker below 5 kg/day · finished goods ageing · truck without gate pass · store dispatched above stock · batch balance difference at closing.

Excel imports — each upload is previewed (new · unchanged · errors), approved, and never duplicated

PHASE	REPORTS
1 Intake	Transfer to Production · 1 RCN Receipts · 2 Input RCN Quality · 19 RCN Receipts Tally Sheet (3 RCN Stock is calculated)
2 Calibration	4 Calibration Process · 5 Calibrated Silos Stock · 24 Wholes Manual Calibration
3 Cooking	6 Cooking Process · Moisture & Temperature Check · 21 Boiler Consumption
4 Shelling	7 Shelling Process · 23 Line-wise · Shelling Frequency · 13 Machine Quality · 11 Shell Waste Sample · 16 Shell Dispatch
5 Colour sort	12 Shelled Kernels Quality
6 Borma	18 Borma Loss Tally · 20 Borma Machine Monitoring · 14 Final Output Analysis (8 Borma Process is calculated)
7 · 8	Manual Peeling Individual Performance · Packing Traceability FG Aging Report · 15 Waste Management Dispatch
No workbook yet	Water Consumption · Humidification & QC-10 · Peeling machine & husk winnowing · QC-12 · QC-9 · N_2 / pouch / carton consumption — forms added in the app
Links	Theoretical Export Yield Prediction (Fiche CCA → production bin) · Daily production report (work-in-progress moves)